Example 1 - Single Doc

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Contents

[1i] Summary 1
[2i] Load Combinations 1
[3v] Loads and Geometry 2
[4v] Beam Response 4

[1i] Summary

This rivt example calculates the maximum stress and deflection in a simply supported, uniformly loaded beam. It also serves as an annotated example of a single doc with multiple sections (a single doc does not use the report generating script).

The example illustrates the use of some of the most common API functions, commands and tags. Further details are provided in the rivt user manual.

The file may be formatted as a text, PDF or HTML doc by changing the type parameter in the PUBLISH command of the Doc API (rv.D) at the end of the file. Published files are found in the respective sub-folders of the _published folder.

[2i] Load Combinations

This is an inline table using the restructured text syntax.

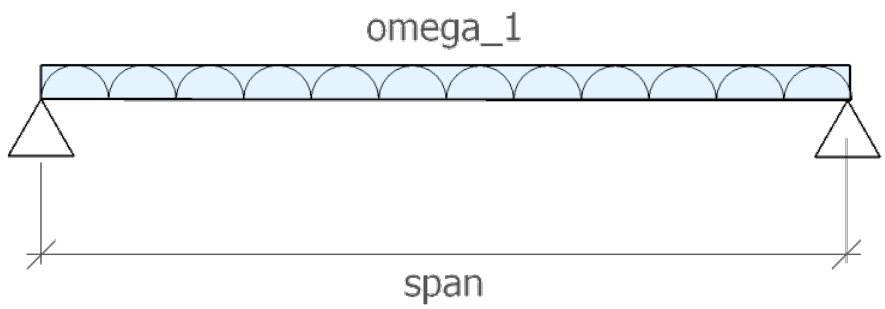
Table 1: ASCE 7-05 Load Effects

Equation No.	Load Combination
16-1	1.4(D+F)
16-2	1.2(D+F+T) + 1.6(L+H) + 0.5(Lr or S or R)
16-3	1.2(D+F+T) + 1.6(Lr or S or R) + (f1L or 0.8W)

When an inline table is contained in a TABLE block it produces the same output as above and also writes the table to a CSV file.

Table 2: ASCE 7-05 Load Effects (saved as csv in _storeed folder)

Equation No.	Load Combination
16-1	1.4(D+F)
16-2	1.2(D+F+T) + 1.6(L+H) + 0.5(Lr or S or R)
16-3	1.2(D+F+T) + 1.6(Lr or S or R) + (f1L or 0.8W)



Maximum Bending Stress

[Eq.1]

$$\sigma_1 = \frac{M_1}{S_1}$$

[3v] Loads and Geometry

Successive lines of value definiiions are formatted as a table. Variable values are defined with the define operator. The line tag [T] labels and numbers the table.

Table 3: Define Unit Loads

variable	value	[value]	description
D_1	3.80 psf	0.18 kPA	joists DL
D_2	2.10 psf	0.10 kPA	plywood DL
D_3	10.00 psf	0.48 kPA	partitions DL
D_4	3.00 klf	43.78 kN_m	fixed machinery DL
L_1	40.00 psf	1.92 kPA	ASCE7-O5 LL

The VALTABLE command reads variable values from the file in the _src folder. The text is used as the table title. The range specifies the starting and ending line to be read from the file (0:0 means all lines). The num;non parameter specifies whether theimported table is numbered.

Table 4: Beam Geometry [file: beam1.csv]

variable	value	[value]	description
spc_1	2.00 ft	0.61 m	beam spacing

variable	value	[value]	description
spn_1	16.00 ft	4.88 m	beam span

Uniform Distributed Loads

[Eq.2]

```
dl_1 = 1.2*(D_4 + spc_1*(D_1 + D_2 + D_3))
```

D_3	D_2	D_4	D_1	spc_1
10.00 psf	2.10 psf	3.00 klf	3.80 psf	2.00 ft

dl_1	[dl_1]	reference
3.64 klf	53.09 kN_m	dead load: ASCE7-05 2.3.2

[Eq.3]

```
ll_1 = 1.6*L_1*spc_1
```

L_1	spc_1
40.00 psf	2.00 ft

ll_1	[ll_1]	reference
0.13 klf	1.87 kN_m	live load: ASCE7-05 2.3.2

[Eq.4]

```
omega_1 = dl_1 + ll_1
```

dl_1	ll_1
3.64 klf	128.00 ft·psf

omega_1	[omega_1]	reference
3.77 klf	54.96 kN_m	total load: ASCE7-05 2.3.2

[4v] Beam Response

Function Table 5: Beam functions **[file:** _src/sectprop.py **]**

Function	Docstring
rectsect(b, d)	section modulus of rectangle
rectinertia(b, d)	moment of inertia of rectangle
midspan_delta(ln, w, e, i)	mid-span deflection of simply supported beam with UDL

[Eq.5]

```
section_1 = rectsect(10*inch, 18*inch)
```

section_1	[section_1]
540.00 in3	8849.01 cm3

[Eq.6]

```
inertia_1 = rectinertia(10*inch, 18*inch)
```

inertia_1	[inertia_1]
4860.0 in4	202288.5 cm4

Bending Stress

[Eq.7]

```
                   2
      omega_1*spn_1
m_1 = --------------
            8
```

omega_1	spn_1
3.77 klf	16.00 ft

m_1	[m_1]	reference
120.52 ftkip	163.40 mkN	mid-span UDL moment

[Eq.8]

```
          m_1
fb_1 = ---------
       section_1
```

section_1	m_1
540.0 inch3	120.5 ft2·klf

fb_1	[fb_1]	reference
2678.2 lb_in2	18.5 MPA	bending stress

[Eq.9]

```
fb_1 < 20000*lb_in2
```

fb_1	20000*lb_in2	ratio	check	reference
2.68 ksi	20.00 ksi	0.133908	OK	stress ratio

Deflection

[Eq.10]

```
delta_1 = midspan_delta(spn_1, omega_1, 29000*ksi, inertia_1)
```

delta_1	[delta_1]
0.04 inch	1.00 mm